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H. M. MOSSMAN ET AL

2,257,063

BLASTING PLUG

Filed Oct. 19, 1940

Fig. 1.

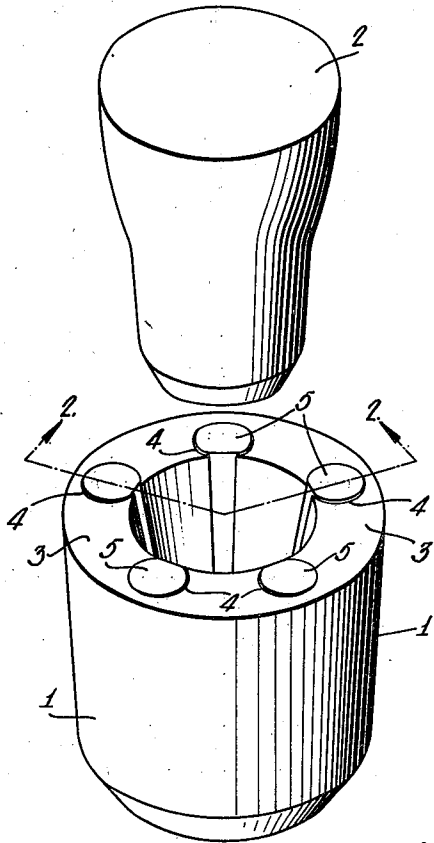


Fig. 6.

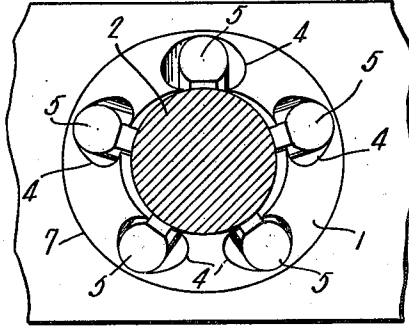


Fig. 2.

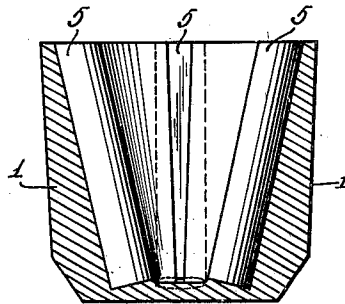


Fig. 3.

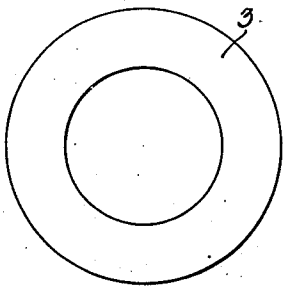


Fig. 4.

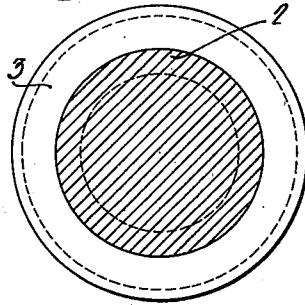
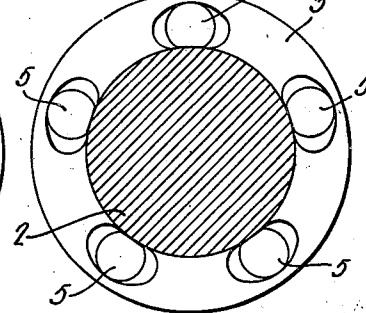


Fig. 5.



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UNITED STATES PATENT OFFICE

2,257,063

BLASTING PLUG

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Application October 19, 1940, Serial No. 361,973

9 Claims. (Cl. 102—11)

This invention relates to improvements in blasting plugs of the type disclosed in U. S. Patent No. 2,007,568.

In plugs of this type wherein the body is composed of rubber or rubber-like material, radial expansion by action of the expander is accompanied necessarily by a stretching of the wall of said body and a corresponding reduction in the effective thickness of the wall. Thus a relative movement of the expander which increases the inside diameter of the body affords a materially lesser increase in the outside diameter of the body. This factor imposes a definite limitation on the ability of the plug to accommodate itself to different diameters of bore hole.

The diametrical size of a bore hole drilled in dirt, rock, or other mineral deposit is not always consistent with the size of the drill. In certain types of rock, for example, the hole may conform accurately to the size of the drill, whereas in other cases, by reason of the physical characteristics of the deposit, the hole may be considerably larger in diameter than the effective diameter of the drill used. Variations also occur in the size of a bore hole due to the wearing away or dulling of the drill, and these variations differ in accordance with the rapidity with which the drill becomes worn. The individual idiosyncrasies of the drillers also may affect the ultimate size of the hole obtained with the use of any given size of drill.

It is desirable, therefore, that a blasting plug shall afford considerable latitude for expansion in order that the plug may accommodate itself to these unavoidable variations in the ultimate size of the bore hole, and a principal object of the present invention is to provide plugs of the type set forth which shall afford this relatively great latitude for expansion.

Another object of the invention is to provide a blasting plug which shall more readily adapt itself to bore holes that are irregular or out of round.

In the attached drawing:

Figure 1 is a view in perspective of a plug made in accordance with our invention, the two principal elements of the plug, namely, the expandible body and the expanding element, being separated one from the other;

Fig. 2 is a section on the line 2—2, Fig. 1;

Figs. 3, 4 and 5 are diagrammatic views illustrating functional aspects of the invention, and Fig. 6 is a transverse sectional view illustrating another characteristic of the device.

With reference to Figs. 1 and 2 of the drawing,

the plug therein illustrated comprises a cup-shaped body member 1, which, preferably, is composed of rubber or rubber composition and which, therefore, possesses the yielding characteristics providing for radial expansion through the medium of a tapered expander 2. Conventionally, the expander 2 is made of wood, although other suitable materials may be used. Departing from the conventional practice, we provide the wall 3 of the body 1 with a plurality of longitudinal channels 4, each of which receives a pin 5 which, preferably, is made of wood or other suitable non-deformable material, or of material which at least has a substantially lesser tendency to deform under strain than the material of the body member 1.

As shown in Fig. 2, the pins 5 will preferably extend to points closely adjacent the bottom or closed end of the body member 1, and preferably also the channels 4 which receive the pins intersect the inner surface of the body member so that the pins 5 are exposed directly to the action of the expander 2. In the illustrated embodiment of the invention, the walls of the body member 1 are of greater thickness towards the bottom of the body member than towards the open top, so that the diameter of the interior of the cup decreases towards the bottom. Thus the pins 5, following the inclination of the inner wall, show a definite convergence towards the bottom of the body member. At the top of the body member 1, the wall thickness preferably is only moderately greater than the effective thickness of the individual pins 5.

The functional characteristics of a plug body made in the aforescribed manner are illustrated in Figs. 3, 4 and 5. Fig. 3 may be considered to be a top end view of a plug body of rubber or like composition and lacking the inserted pins described above. Let it be assumed that the inside diameter of this member is expanded, through the medium of the inserted expander 2, to a predetermined extent such, for example, as shown in Fig. 4. As explained above, this expansion of the inner diameter will effect an expansion of the outside diameter, but to a lesser radial extent, the effective thickness of the wall 3 decreasing as illustrated. To better illustrate this effect, the normal annulus shown in Fig. 3 is superimposed in broken lines upon the expanded annulus shown in Fig. 4. It will be apparent that the increase in outside diameter is materially less than the increase in the inside diameter of the annulus.

In Fig. 5, we have illustrated the effect of ex-

pansion, to the same extent, of a plug body made in accordance with the present invention. It will be noted that in this case the increase in effective outside diameter is only slightly less than the increase in inside diameter, this being due to the fact that a minimum wall thickness is established by the presence in the wall of the pins 5. It is apparent, therefore, that a plug made in accordance with the present invention will accommodate itself effectively to bore holes of substantially greater diameter than will a plug of the same normal dimensions but lacking provision of the pins 5.

It is to be noted that when the expander 2 is forced into the body member 1, the initial tendency is for the pins 5 to move laterally, while more or less retaining their normal inclined positions with respect to each other. If, however, the upper portion of the body member is constrained by contact with the walls of the bore hole from further expansion, the tendency of the pins will be to pivot around their upper ends, the lower or inner ends of the pins being forced outwardly away from the center of the cup and toward the wall of the bore hole. This pivotal movement of the pins will have the effect of bringing the lower or bottom end of the cup-shaped body member 1 into more intimate contact with the walls of the drilled hole, and to thereby increase the area of engagement between the body member 1 with the opposed wall of the bore hole in which it is established.

Another desirable characteristic of a plug made in accordance with this invention resides in its ability to conform or adapt itself more readily to bore holes that are irregular or out of round. This is illustrated in Fig. 6 wherein the plug is shown expanded in a hole of somewhat oval shape. This ability of the plug to conform readily to holes other than round is due to the action of the inserted pins described above. As the expander 2 is driven into the body 1 of the plug, the expansion of the latter occurs more or less uniformly to the point where the wall of the plug is solidly engaged with the narrower dimension of the bore hole. During this initial expansion, the pins 5 are forced outwardly more or less in their normal inclined positions. When, however, the upper ends of certain of the pins are solidly backed by the wall of the bore hole, further movement of the expander results in the aforescribed pivotal movement of these pins around their upper ends, whereas the pins adjoining the greater dimension of the hole continue to move outward in the normal manner. The plug is thereby made to assume a shape corresponding to the shape of the bore hole and is solidly clamped in this form and around the entire periphery against the wall of the hole.

While various shapes of expander element may be used, it is preferred, in order that the advantages described above may be realized in maximum degree, to form the expander, at least in the leading portion, with a taper of lesser included angle than the normal angle of convergence of the pins 5. The expander will then initially engage the pins 5 at points somewhat below the top of the plug body, and will afford more latitude for self-adjustment of the pins under the expander action. A preferred form of expander is shown in Fig. 1, this form having been found to afford the best results under widely varying conditions.

We claim:

1. A blasting plug comprising a substantially

cylindrical hollow body of rubber or like deformable material adapted for radial expansion under internally-applied pressure, said body having in the wall thereof a circumferentially arranged series of elements relatively resistant to deformation and extending longitudinally in said body.

2. A blasting plug comprising a substantially cylindrical hollow body of rubber or like deformable material adapted for radial expansion under internally-applied pressure, said body having in the wall thereof a circumferentially arranged series of elements relatively resistant to deformation, said elements extending longitudinally in said body and individually embracing a substantial part of the radial thickness of said wall.

3. A blasting plug comprising a substantially cylindrical hollow body of rubber or like deformable material adapted for radial expansion under internally-applied pressure, said body having in the wall thereof a circumferentially arranged series of rods extending longitudinally of the body, said rods being composed of material relatively resistant to deformation as compared with the material of said wall and being sufficiently thick to embrace a substantial portion of the radial thickness of the latter.

4. A blasting plug comprising a substantially cylindrical hollow body of rubber or like deformable material adapted for radial expansion under internally-applied pressure, said body having in the wall thereof a circumferentially arranged series of substantially rigid rods extending longitudinally of the body and having a thickness not materially less than the thickness of said wall.

5. A blasting plug comprising a substantially cylindrical body of rubber or like deformable material having an axial recess in an end thereof, said body being adapted for radial expansion through the medium of an expander forcibly entered in said recess, and means relatively resistant to deformation as compared with said body and established in the wall embracing said recess for limiting reduction in the effective thickness of said wall under the stretching effect of said expander.

6. A blasting plug comprising a substantially cylindrical body of rubber or like deformable material having an axial recess in an end thereof, said body being adapted for radial expansion through the medium of an expander forcibly entered in said recess, and a circumferentially arranged series of relatively non-deformable elements established in the wall embracing said recess and extending longitudinally of said cylindrical body.

7. A blasting plug comprising a substantially cylindrical body of rubber or like deformable material having an axial recess in an end thereof, said body being adapted for radial expansion through the medium of an expander forcibly entered in said recess, the wall embracing said recess having a circumferentially arranged series of channels extending longitudinally of said body and intersecting the inner surface of said wall, and a corresponding series of rod-like elements inserted in said channels, said elements being composed of a material relatively non-deformable as compared with the deformable material of said plug.

8. A blasting plug comprising a substantially cylindrical body of rubber or like deformable material having an axial recess in one end there-

of, said body being adapted for radial expansion through the medium of an expander forcibly entered in said recess, the inner surface of the wall embracing said recess converging inwardly of said body and having a circumferentially arranged series of channels extending longitudinally of the body and substantially parallel with the said converging inner surface, and a corresponding number of rod-like elements inserted in said channels, said elements being composed of a material relatively non-deformable as compared with the deformable material of said body.

9. A blasting plug comprising a substantially cylindrical body of rubber or like deformable material having an axial recess in an end there-

of, said body being adapted for radial expansion through the medium of an expander forcibly entered in said recess, the inner surface of the wall embracing said recess converging inwardly and angularly with respect to the outer surface whereby the wall increases in radial thickness towards the bottom of said recess, and a circumferentially arranged series of relatively non-deformable elements established in said wall and extending longitudinally of said body, said elements substantially paralleling the inner surface of said wall and having a thickness not materially less than the radial thickness of the wall at its outer end.

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